

Public Chat Portal for Alkhidmat

KAAVISH PROJECT PROPOSAL

By

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1 Abstract

The proposed project aims to design and implement a public-friendly chat portal for the Alkhidmat Foundation Pakistan, one of the country's leading non-profit organizations serving vulnerable communities. This chat portal will be integrated into Alkhidmat's official website and serve as a central communication tool for donors, volunteers, and beneficiaries, enabling them to seamlessly access information across all 12 domains, such as health care (hospitals, diagnostics, blood donations, pharmacies), disaster management (flood relief), community services (mayyat bus, oxygen cylinders), water and sanitation (WASH filtration plants), education (scholarships, madaris), Islamic loans (interest-free Qarza), orphan care (education, health screenings, Agosh homes), Bano Qaabil (free IT courses), youth and volunteer programs, and women empowerment.

Currently, inquiries are handled manually via phone calls and emails, which places a heavy burden on staff and slows down response times. Navigating the website can also be confusing for users with limited digital literacy. This project will address these challenges by introducing an AI-powered conversational interface that automates responses to common questions, while also escalating complex or sensitive queries to human staff through an integrated dashboard.

The system will utilize retrieval-augmented generation (RAG) to provide rapid, accurate responses drawn from Alkhidmat's public resources, such as FAQs, schedules, and service descriptions. It will also incorporate an intent classification module to categorize user queries into domains like health or donations, ensuring tailored responses. Crucially, the chat portal will support multimodal input, allowing users to communicate via text, voice message (which will be converted to text by the LLM), and images (which the LLM will process for context). This will significantly enhance accessibility. The chat portal will feature an escalation mechanism to hand off unresolved queries to the appropriate staff, guaranteeing that no user request is left unanswered. Multilingual support for English, Urdu, and Roman Urdu will enhance accessibility, with agentic AI handling translation and intent confirmation for precise interactions.

Beyond automation, the project emphasizes human-computer collaboration. While AI provides first-line support, human agents will handle nuanced interactions through a dedicated staff dashboard, streamlining workload and improving service efficiency. Additionally, an analytics module will log and evaluate user interactions, measuring performance through response times, escalation rates, and query trends. These insights will help Alkhidmat identify gaps in service delivery and better understand community needs.

Socially, this initiative directly contributes to inclusive humanitarian service delivery in Pakistan. Many underserved communities face barriers such as a lack of awareness, limited digital literacy, and delayed communication. A 24/7 multimodal AI-assisted chat portal provides users with an accessible, self-service channel that reduces friction and empowers them to engage more actively with Alkhidmat's programs. Meanwhile, the automation of repetitive queries will free up staff resources, enabling Alkhidmat to concentrate on field operations and strategic initiatives.

The originality of this system lies in its multimodal accessibility, domain-aware classification, and context-sensitive escalation, which distinguish it from generic chatbot solutions deployed by some organizations. It is designed specifically around Alkhidmat's operational model, making it both practical and impactful. This balance of innovation and social good makes the project a scalable model for non-profits worldwide.

The project integrates knowledge from multiple areas of computer science, including natural language processing (NLP), machine learning, human-computer interaction (HCI), information retrieval, computer vision, and software engineering. The one-year timeline enables iterative devel-

opment across the design, prototyping, implementation, and testing phases. The final outcome will be a live prototype deployed on a cloud platform, along with documentation to ensure scalability and maintainability.

In summary, this project combines advanced AI techniques with human-centered design to address a pressing real-world challenge: enhancing accessibility, efficiency, and transparency in humanitarian services. By delivering an intelligent, intuitive, and socially impactful communication system, the chat portal will not only enhance Alkhidmat's operations but also stand as an example of how AI for social good can empower communities.

2 Problem definition

Alkhidmat Foundation Pakistan, a prominent non-profit, receives a high volume of inquiries daily from donors, volunteers, and beneficiaries seeking information on donations, events, volunteering, or health camps. At present, these queries are managed through manual channels such as phone calls and emails, which is time-intensive, resource-heavy, and inefficient. Staff members often face repetitive queries that could be automated, while users encounter delays and difficulties in accessing information.

The absence of a dedicated chat solution also limits engagement, especially for individuals with limited digital literacy who cannot easily navigate static website pages. The proposed AI-powered chat portal solves this problem by providing instant automated responses for routine queries and intelligent routing of complex cases to staff, ensuring both efficiency and inclusivity. By accepting voice messages and images, it lowers the barrier to entry, especially for users who prefer speaking over typing or need to share visual information (e.g., a photo of a location for disaster relief, or a medical report).

3 Social relevance

The project addresses critical societal barriers to equitable access to humanitarian services and the operational challenges faced by non-profit organizations. In Pakistan, where Alkhidmat operates extensively, underserved communities often depend on timely access to disaster relief, medical camps, and welfare initiatives. However, obstacles such as limited digital literacy, language diversity, and delayed communication can prevent individuals from receiving the help they need. By providing a 24/7 multilingual multimodal AI-powered chat interface, the system enables users to easily self-serve information, even outside working hours, thereby enhancing inclusivity and engagement. The option to use voice-to-text and image upload directly accommodates users with low digital literacy or those who are not comfortable with typing in a specific language. At the same time, it reduces the repetitive workload on staff, allowing them to redirect efforts toward high-impact fieldwork and service delivery. This dual impact strengthens transparency, trust, and user satisfaction, while amplifying Alkhidmat's ability to serve communities more effectively and in alignment with its public service mission.

4 Originality/Novelty

Unlike generic chatbot solutions that many organizations use, this system is customized to Alkhidmat's operational needs. Its unique features include:

- **Multimodal Input Processing:** Integrating voice-to-text for voice messages and image processing/analysis (using a Multimodal LLM or separate models) into the core RAG workflow. This provides a user-friendly and inclusive way for users to submit queries, such as sending a voice message with a question or an image related to a problem (e.g., a damaged water pump, a flooded street).
- **Domain-specific classification** of queries (health, donations, volunteering, events), ensuring that responses are accurate and contextually relevant.
- **Context-aware escalation**, when the AI cannot resolve a query with confidence, the system intelligently routes the conversation to the appropriate human team member, avoiding dead ends or generic fallback messages.
- **Integrated analytics for Continuous Improvement:** Beyond handling queries, the system incorporates a detailed analytics module that tracks community engagement and highlights unmet needs. For example, it can measure how long it takes for a query to be answered, which domains consistently require longer response times, and how quickly users receive escalated answers from staff. These insights not only improve user experience but also help Alkhidmat identify bottlenecks, allocate staff more effectively, and refine its services over time.

This tailored approach combines cutting-edge AI techniques with practical usability, making it a pioneering model for AI-driven non-profit service delivery.

5 CS contribution

The project involves multiple core areas of computer science:

- **Natural Language Processing (NLP):** Intent classification, text understanding, and retrieval-augmented generation (RAG) system. Includes Speech-to-Text for voice message transcription.
- **Information Retrieval:** Efficient extraction of answers from structured and unstructured data sources.
- **Machine Learning & Deep Learning:** Training lightweight classifiers and applying neural network-based models to improve accuracy and contextual understanding. This now includes models for multimodal understanding (processing images in a chat context).
- **Computer Vision (CV):** Techniques for interpreting and extracting information from uploaded images (e.g., OCR, object recognition, scene understanding) to supplement text-based queries.
- **Computational Intelligence:** Leveraging optimization and learning techniques for classification, escalation logic, and system adaptation.
- **Database Management Systems:** Structuring and querying service data for efficient retrieval, ensuring reliability and scalability.
- **Web and App Development:** Designing and implementing the frontend chat interface, backend APIs, and the human-agent dashboard for seamless usability.

- **Human–Computer Interaction (HCI):** Developing user flows, UX wireframes, and intuitive interaction design for diverse stakeholders. This specifically includes the design of the voice message and image upload features for maximum accessibility.
- **Data Science & Analytics:** Logging, tracking performance, analyzing query patterns, and providing actionable insights to improve system quality.

These contributions draw directly from advanced coursework the team has completed – including Deep Learning, Computational Intelligence, Database Management Systems, Web and App Development, and Data Science & Analytics – bridging theory with practical real-world deployment.

6 Scope and Deliverables

The project’s scope is well-suited for a four-member team over a year, balancing ambition with feasibility. The first semester will focus on planning, user flows, data integration, and prototyping; the second on implementation, testing, and refinement. This timeline supports iterative development and user feedback.

Phase 1 (Fall 2025):

- Requirement gathering, user flow & domain mapping
- UX wireframes and chat interface design
- Data collection, cleaning, and consolidation
- Initial prototype of the retrieval and classification modules and integration of a voice-to-text API/library.

Phase 2 (Spring 2026):

- Integration of RAG, classifier, and chatbot
- Integration and testing of image processing logic/multimodal LLM capability.
- Escalation routing to staff dashboard
- Logging, analytics, and performance evaluation
- Cloud deployment and documentation

Deliverables:

- **User Flow & Domain Map:** Flowchart of user interactions and routing with voice and image input support.
- **UX Wireframes/Chat Screens:** Interface sketches
- **RAG System Setup:** Retrieval of answers from public data
- **Intent Classifier:** Categorization of queries by domain
- **Multimodal Input Module:** Functionality for processing and converting voice messages to text and analyzing uploaded images for context.
- **Chatbot Module:** Web interface with AI responses and escalation

- **Human Agent Dashboard:** Staff interface for escalated queries
- **Logging & Analytics:** Tracking queries, response times, and escalations
- **Documentation & Codebase:** README, user guide, setup instructions

These deliverables align with the objectives and remain achievable within the team's capacity.

7 Feasibility

Datasets:

The chatbot will primarily rely on resources, and data from Alkhidmat Foundation, including FAQs, program details, event schedules, donor guidelines, and health camp announcements. These datasets can be collected and cleaned directly from Alkhidmat's website, reports, and outreach materials. Additional conversational datasets (e.g., from open-source question, answering corpora) may be used for pre-training and fine-tuning intent classification. Since the majority of required data is already available and accessible, data availability is not a bottleneck. For multimodal input, we will utilize established third-party APIs (e.g., Google Cloud Speech-to-Text, pre-trained multimodal models) to handle the complex processing, minimizing the need for large, custom-labeled datasets for voice and image understanding.

Compute Resources:

Since the project does not demand training large-scale AI models from scratch, its compute requirements are moderate. Open-source pre-trained models like Llama 2 (or a LLaVA-style model for image processing) can be fine-tuned or directly applied through frameworks such as LangChain. Training and inference can be handled using standard university lab machines or cloud-based GPU/CPU services on an as-needed basis. For deployment, lightweight containerized environments (via Docker) hosted on platforms such as Heroku or Netlify will ensure scalability without heavy infrastructure costs. The use of external APIs for real-time Speech-to-Text and Multimodal LLM calls will offload significant compute demands from the host server.

Software Libraries and Tools:

The project builds upon mature, open-source ecosystems. Python-based libraries (FastAPI for backend APIs, LangChain for retrieval-augmented generation, scikit-learn/PyTorch for intent classification, and Elasticsearch for indexing) provide reliable support. React and CSS frameworks will be used for the front-end chat interface and dashboard. All libraries are free and well-documented, reducing technical risks and ensuring quick development. Docker will allow consistent development-to-deployment transitions. For voice-to-text, we will investigate reliable APIs (e.g., Google, Azure) or open-source libraries (e.g., Whisper) for transcription, and for image processing, we will use a multimodal component within LangChain or a dedicated library like OpenCV for basic pre-processing before LLM analysis.

Hardware Resources

The team's personal laptops (upto 16GB RAM) and university computing facilities will be sufficient for development, prototyping, and testing. No specialized hardware is required beyond access to occasional cloud services for hosting or model fine-tuning. This minimizes hardware costs and ensures the project remains low-budget and sustainable.

In summary, the project is highly feasible due to the availability of public datasets, reliance on open-source tools, modest hardware requirements, and the strong alignment between the technical plan and team expertise. This balance of resources, costs, and skills ensures that the proposed chat portal can be successfully developed, deployed, and sustained within the scope of a year-long academic project.

8 Team dynamics

Fatima leads AI, data analysis, and front-end work, using knowledge from courses like Computer Vision, Reinforcement Learning, Probabilistic Graphical Models, Data Science, and Intro to Data Analytics. Her internship as a Business Intelligence Intern at Waada improved her skills in handling data queries, and her work on the DonorNet project (a tool for managing donors with AI) helps shape the chat portal's query and escalation features. A possible challenge is that she has limited experience with long-term project management, which may make balancing deadlines difficult; however, her strong technical foundation positions her well to deliver her contributions on time with team support. Her background in Computer Vision will be essential for integrating and testing the image processing capabilities.

Yashfeen strengthens our machine learning efforts, especially for sorting user questions into categories. She worked on DonorNet and a Diabetic Retinopathy Detection project, using deep learning for medical data, which connects to the portal's answer retrieval and question classification. Her courses in Data Science, Competitive Programming, Deep Learning (ongoing), equip her to improve the system's accuracy and speed, including optimizing the voice-to-text and image analysis outputs for the RAG system.

Rubab brings research experience from LUMS on large-scale biological data analysis and web systems, making her well-suited to build the analytics module for tracking user queries. Her Data Structures II project combined pattern recognition with front-end design, showing her ability to merge insights with usability. Currently taking a Web Development course, she adds valuable frontend and backend skills to ensure the portal is data-driven, adaptive, and user-friendly. A challenge for her is inexperience with deployment processes, which will require guidance during the cloud hosting and integration phase; however, her adaptability and research background help offset this along with Samah's guidance. She will focus on the front-end implementation of the voice and image upload features to ensure a seamless HCI experience.

Samah focuses on backend development, with experience in computational intelligence, data science, and interactive dashboards. She contributed to DonorNet by handling backend workflows and data integration, directly relevant to the chat portal's logging and escalation features. Her skills in database design and Python workflows support analytics and monitoring, while her Power BI and Python dashboard experience strengthens reporting. Coursework in API development, Docker, and DevOps practices provides practical expertise for deploying the system with CI/CD pipelines. Her main limitation is limited frontend experience, so she will concentrate on backend and deployment while collaborating with Rubab and Fatima on interface tasks. She will be responsible for integrating the Speech-to-Text and Multimodal processing APIs into the FastAPI backend.

9 Tech stack

Backend: FastAPI (Python) for fast, asynchronous API handling.

LLM/RAG: LangChain with Llama 2 for efficient, context-aware query responses.

Multimodal/Speech: External API or open-source library (e.g., Whisper, Google Speech-to-Text) for Voice-to-Text transcription. Multimodal LLM (e.g., LLaVA, Gemini API, or an image-captioning model) for image analysis.

Database/Search Index: Elasticsearch for quick, scalable data retrieval.

Frontend: React with a Chat UI component and CSS for a responsive, user-friendly interface.

Agent Interface: Custom React-Flask dashboard for staff to handle escalated queries.

Deployment: Docker for consistent environments and Heroku for scalable cloud hosting.

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Undertaking of Kaavish advisement as an External Supervisor

I hereby affirm that I have read the project details as described on the preceding pages and agree to undertake advisement of this Kaavish project as an External Supervisor. I understand that this role entails the following.

Meeting Meeting the project team regularly, at least once every two weeks, for the entire duration of the Kaavish. The meetings may be held remotely if required.

Advisement Providing supervision and advice to the team in order to ensure steady progress of the project toward its goals.

Liaison Liaising with the Internal Supervisor as required, e.g. to provide feedback or engage in grading.

Other Any other task, depending on availability and suitability, relevant to the Kaavish as communicated by the Internal Supervisor or Kaavish Working Group.

Name: _____

Email: _____

Phone: _____

Designation: _____

Affiliation: _____

Signature: _____